## Investigation: SPRINGER SA (CVM code 10960), fiscal year 2013, as it was on 2016-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

## Document used

2013 DFP, version 1, filed on 2014-04-02; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2014-04-02).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=36045

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.139 (index)
- Alert by the study 1 rule: yes
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 97 (n = 16)
- Percentile in group and year: 92 (n = 364)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.720 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 21 (n = 21)
- Percentile in group and year: 15 (n = 459)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.244 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 89 (n = 22)
- Percentile in group and year: 90 (n = 477)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 2.135 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 93 (n = 23)
- Percentile in group and year: 97 (n = 473)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.339 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 70 (n = 22)

Percentile in group and year: 86 (n = 479)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.890 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 15 (n = 17)
Percentile in group and year: 28 (n = 387)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.938 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 61 (n = 22)
Percentile in group and year: 34 (n = 470)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.132 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 98 (n = 23)
Percentile in group and year: 96 (n = 500)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.724 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 2 (n = 23)
Percentile in group and year: 7 (n = 494)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −0.1 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 26.3 days)
Percentile in sector and year: 40 (n = 21)
Percentile in group and year: 49 (n = 446)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 45.69 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 85 (n = 23)
Percentile in group and year: 92 (n = 443)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.120 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 93 (n = 23)
Percentile in group and year: 94 (n = 499)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 6.885 (index)
Components: X1 = 0.338; X2 = 0.168; X3 = −0.135; X4 = 4.788 (indices)
Percentile in sector and year: 94 (n = 24)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage: 0
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
- No long-term debt in either year: the change in leverage is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

## Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

## What was known on the date

- DFP 2010, v1: 2011-03-31
- DFP 2010, v2: 2011-04-19
- DFP 2011, v1: 2012-04-02
- FCA 2013, v1: 2013-03-11
- DFP 2012, v1: 2013-03-27
- FCA 2013, v2: 2013-05-24

- DFP 2013, v1: 2014-04-02
- DFP 2014, v1: 2015-03-31
DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 6 material fact documents and 9 market notice documents.
Material facts filed in the 24 months before the date (from 2014-05-01 to 2016-04-30):
- 2016-03-22: Venda da totalidade das ações que possui da Controlada Springer Plásticos da Amazônia S.A.
- 2015-10-29: Divulgação de procedimentos e cronograma para enquadrar cotações de ações
- 2015-10-23: subject omitted (possible person's name); see the document at the CVM
- 2015-04-13: Venda de participação societária em controlada
Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da SPRINGER S/A. em 31 de dezembro de 2013, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2014 FCA: Confidor - Auditores Associados.
Auditor in the 2015 FCA: Confidor - Auditores Associados, Peppe Associados - Consultores & Auditores Independentes.
Auditor in the 2016 FCA: Peppe Associados - Consultores & Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2015: Confidor - Auditores Associados -> Peppe Associados - Consultores & Auditores Independentes (transition: two firms in the 2015 FCA). The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 22 held-out cases, a single measurement with the frozen reader (dd688da): precision 0.973, coverage 0.915, literal check on 146 of 146 excerpts, no natural-person name (docs/leitor_medida_f1_2.json).

Explanatory notes · Related parties
The reader found no note with "partes relacionadas" (related parties) in the title.

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- [emphasis of matter] “Ênfase Conforme descrito na nota explicativa 3.g, as demonstrações contábeis individuais foram elaboradas de acordo com as práticas contábeis adotadas no Brasil. No caso da SPRINGER S/A. essas práticas diferem da IFRS, aplicável às demonstrações contábeis separadas, somente no que se refere à avaliação dos investimentos em coligadas e controladas em conjunto pelo método de equivalência patrimonial, enquanto que para fins de IFRS seria custo ou valor justo, todavia não houve diferença entre as duas práticas. Nossa opinião não está ressalvada em função desse assunto.” (original excerpt in Portuguese)
- Excerpt omitted; see the auditor's report in the document

Reference Form
Form of 2014, version 2, filed on 2015-05-12.
The Reference Form lists no related-party transactions in the table. It is the only version available in the open data; earlier versions are not there.

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, a header line (years, column titles) may still come out as an excerpt: in the reader's measurement, 1 of 19 header texts of the held-out cases came out as excerpts.

## How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits
- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine
motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-2 (commit dd688da). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: SPRINGER SA.